Deep Learning

Object Detection and Tracking

Outline

Introduction

Evolution of Object Detection Technologies

Deep Learning Era

- R-CNN
- Single Shot Detectors

YOLO (You Only Look Once) Family

- YOLOv1 to YOLOv3 (2016–2018)
- YOLOv4 and YOLOv5 (2020)
- YOLOv6, YOLOv7, YOLOv8 (2022–2024)

Outline

Object Tracking Technologies

- ByteTrack
- DeepSORT

SAM (Segment Anything Model)

- SAM v1 (2023)
- SAM v2 (2024–2025)

YOLO v11/v12 All-In-One

Code Walking (YOLO, YOLO + Tracking, SAM, YOLO Sagmentation)

Introduction

Object Detection

Identifying and **localizing** objects in Images.

Object Tracking

Maintaining object Identity across frames once detected.

Introduction

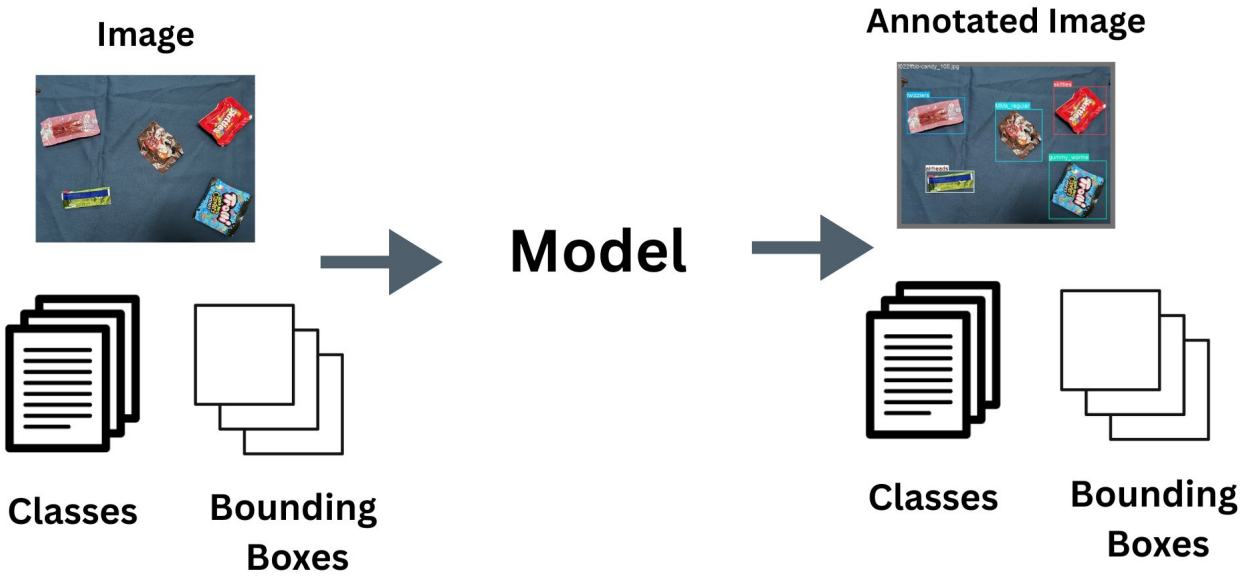
Bounding Box (BBx) Detection

Object Localization by enclosing targets within **Rectangular Boxes**.

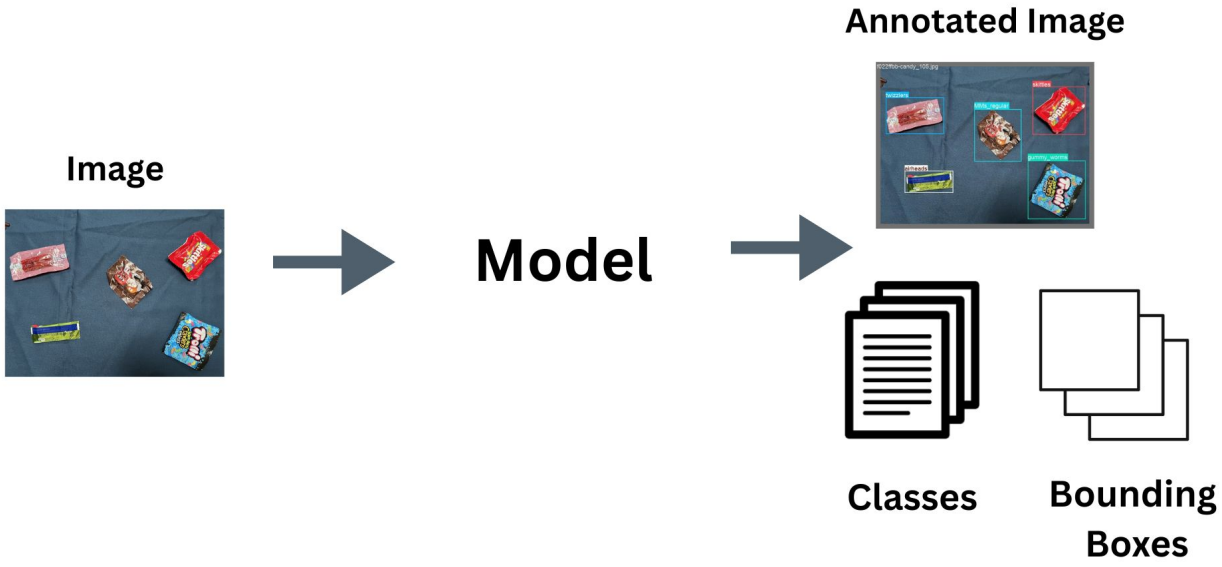
Segmentation

Pixel-Level Delineation of object boundaries for fine-grained understanding.

Introduction



Introduction



Evolution of Object Detection Technologies

Approaches:

- **HOG** (Histogram of Oriented Gradients)
- **SIFT** (Scale-Invariant Feature Transform)
- **Haar Cascades** (Viola–Jones)

Limitations:

- Handcrafted features → Poor scalability
- Sensitive to illumination, occlusion, and perspective variations

Deep Learning Era

R-CNN

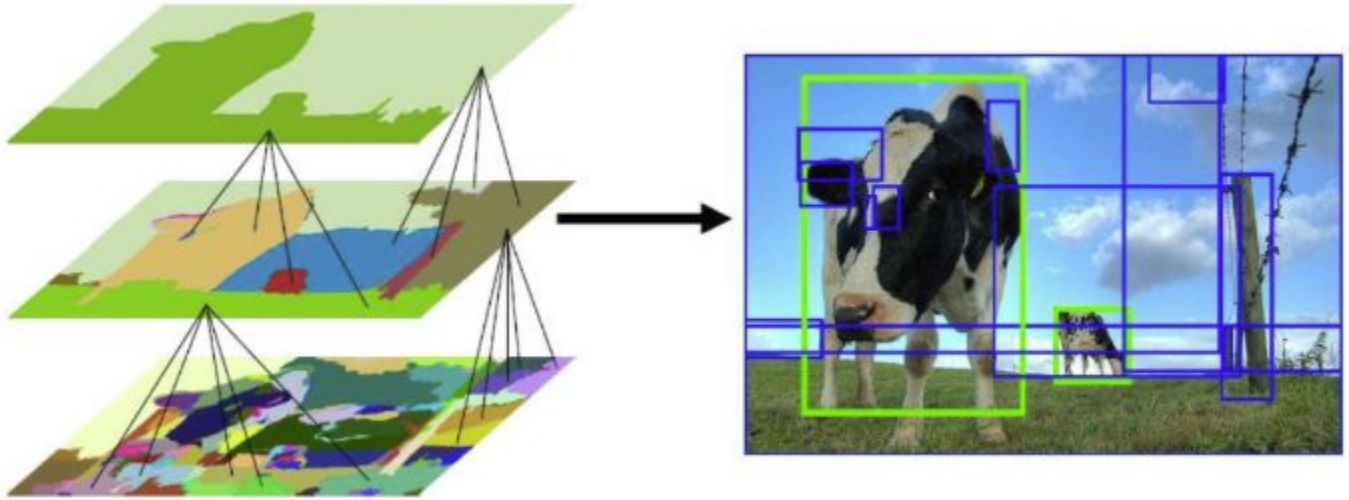
Region-Based Detectors

Proposal Region of Interest based on Feature Descriptor

Selective Search Generate Segment + Hierarchical Grouping

Classify that Region to a Class + Bounding Box Generation

Deep Learning Era



Deep Learning Era

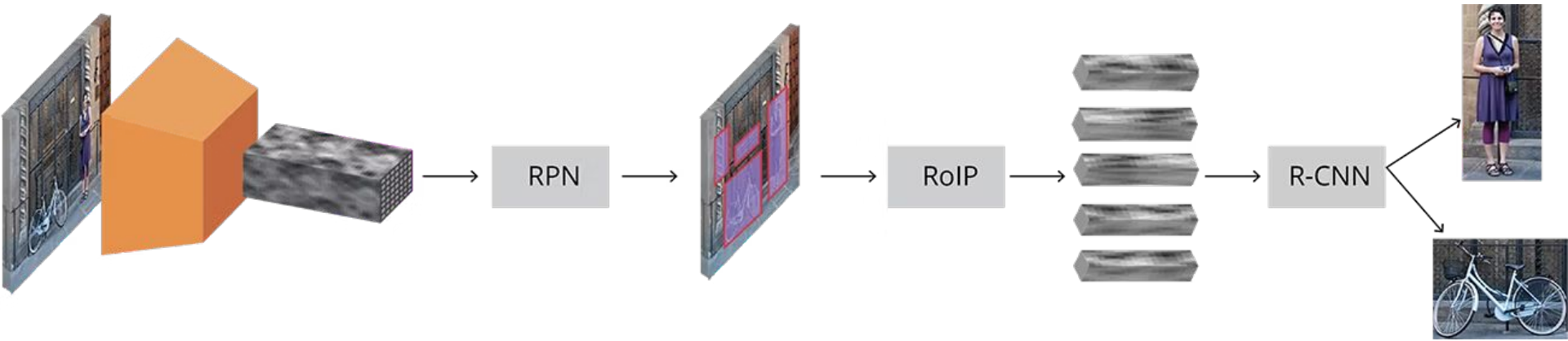
Fast R-CNN

Region Proposal Network

Base CNN + RPN + RP + ROI + R-CNN

RPN uses **Non-Maximum Suppression** Based on IOU

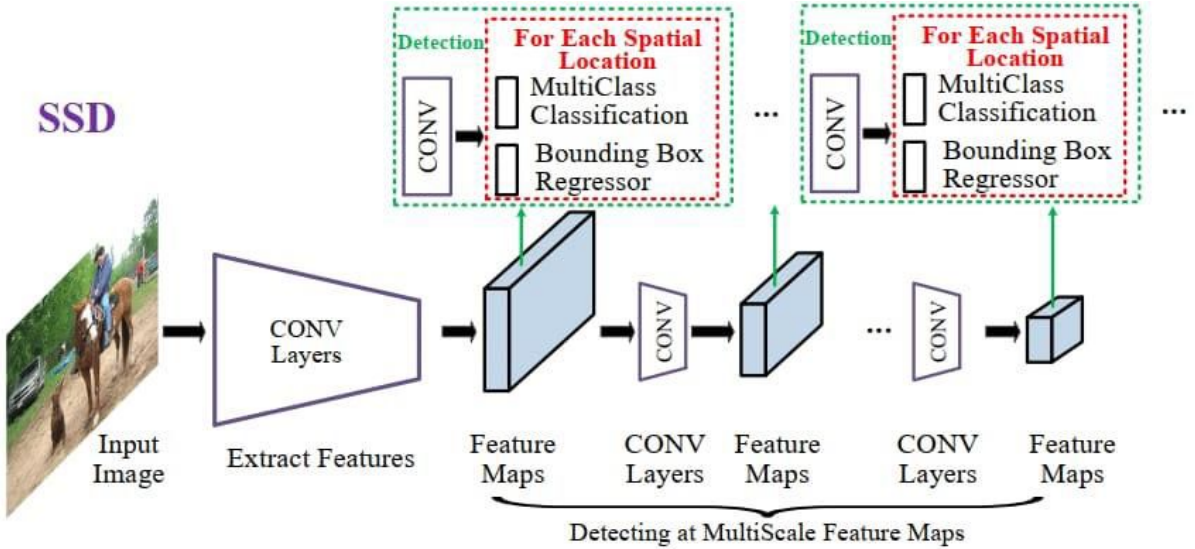
Deep Learning Era



Deep Learning Era

SSD (Single Shot MultiBox Detector, 2016)

Performs **Object Detection** in a **Single Forward Pass** using multiple feature maps from different layers of the network hence the name **Single Shot**.



Deep Learning Era

YOLO (You Only Look Once, 2016)

Backbone **Neck** **Head**

Treats detection as a

Single Regression Task

Dividing the image into an S×S grid where each cell directly predicts **Bounding Boxes**

Objectness Scores

Class Probabilities in one Unified, End-to-End Pipeline

YOLOv1 to YOLOv3 (2016 - 2018)

YOLO

GoogLeNet inspired CNN (24 Conv + 2 FC Layers)

DarkNet-19 as Backbone (19 conv + 5 MP layers)

Multi-Scale Training → Robust to different Image Sizes

Darknet-53 (53 conv layers with **Residual** Connections)

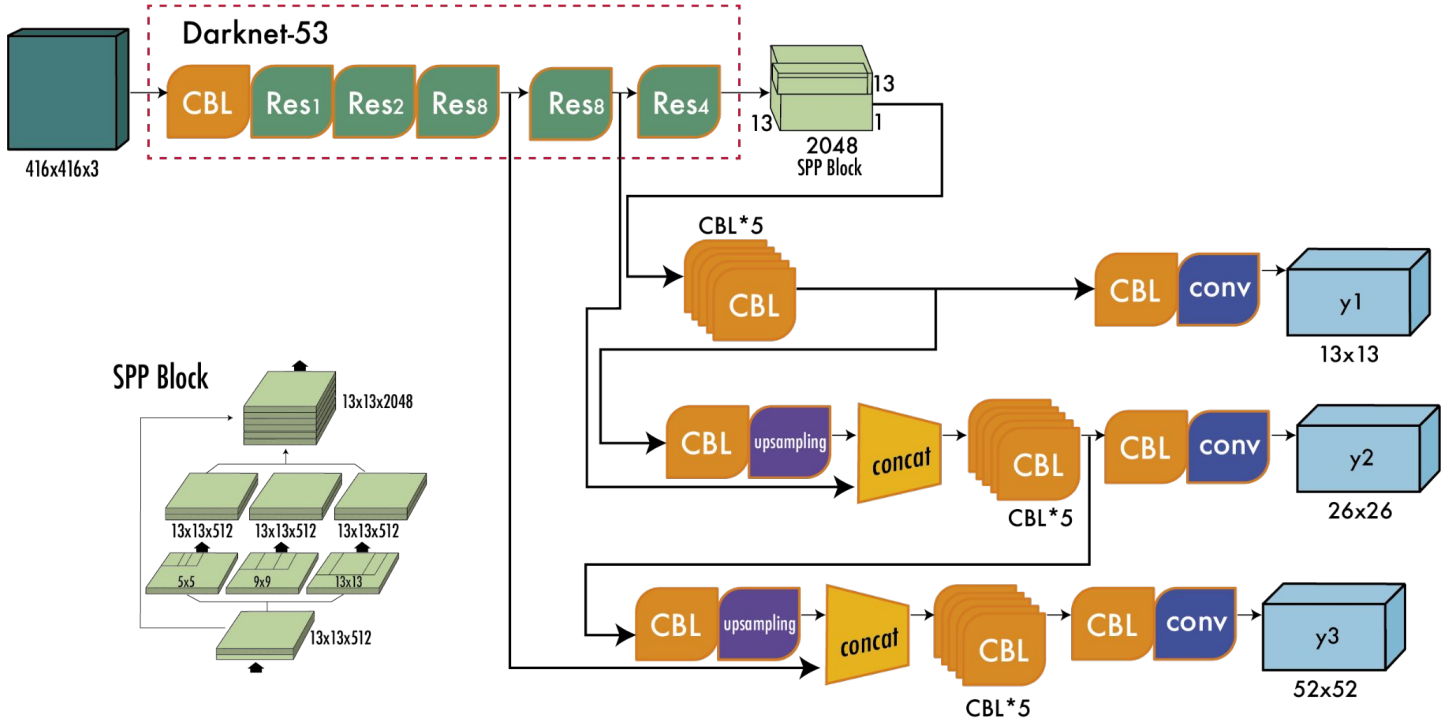
Logistic Classifiers instead of **Softmax** → allows Multi-Label Prediction

SPSS Block **Capture Features at Multiple Receptive Field Sizes**

Mish Activation => $\mathbf{x \times \tanh(\ln(1 + e^{x}))}$

YOLOv1 to YOLOv3 (2016 - 2018)

v3



YOLOv4 to YOLOv7 (2020–2021)

YOLO

CSPDarknet Backbone Cross Stage Partial Connections

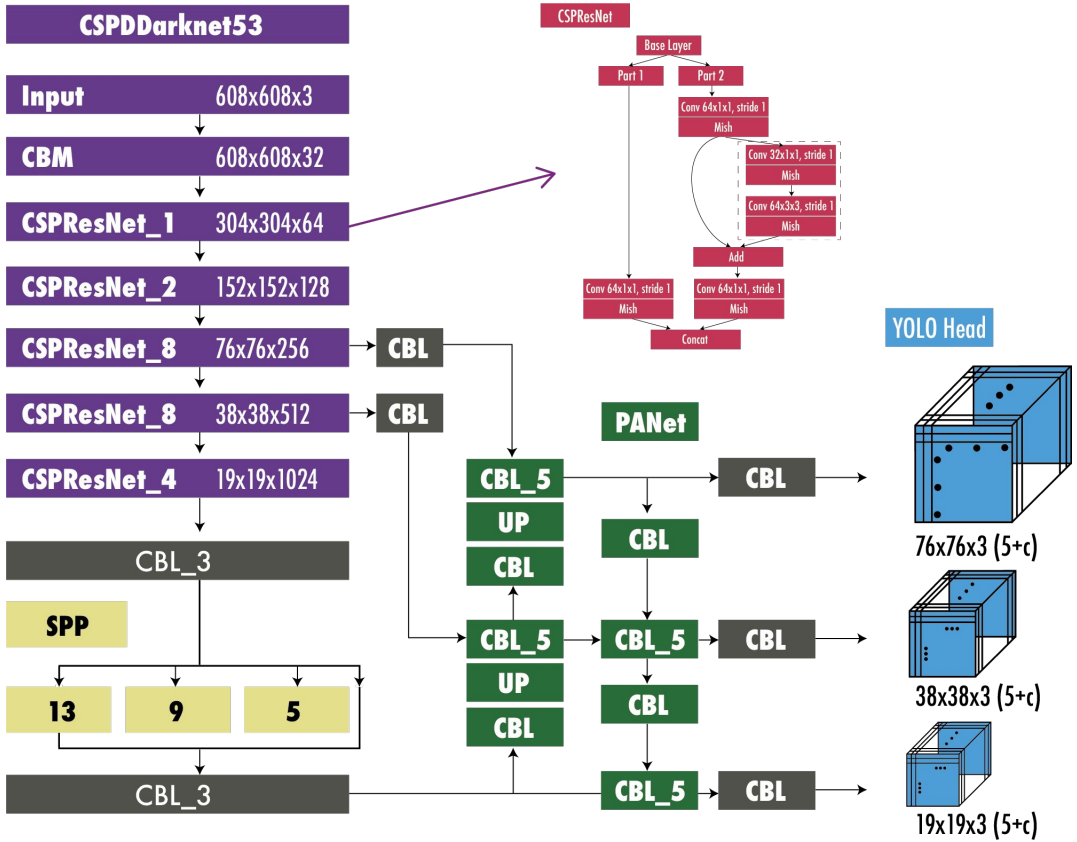
PANet Path Aggregation Combines Bottom-Up and Top-Down Pathways

Advanced Augmentations (**Mosaic**, **CutMix**)

Complete IoU/ **Distance** IoU Loss for better Bounding Box Regression

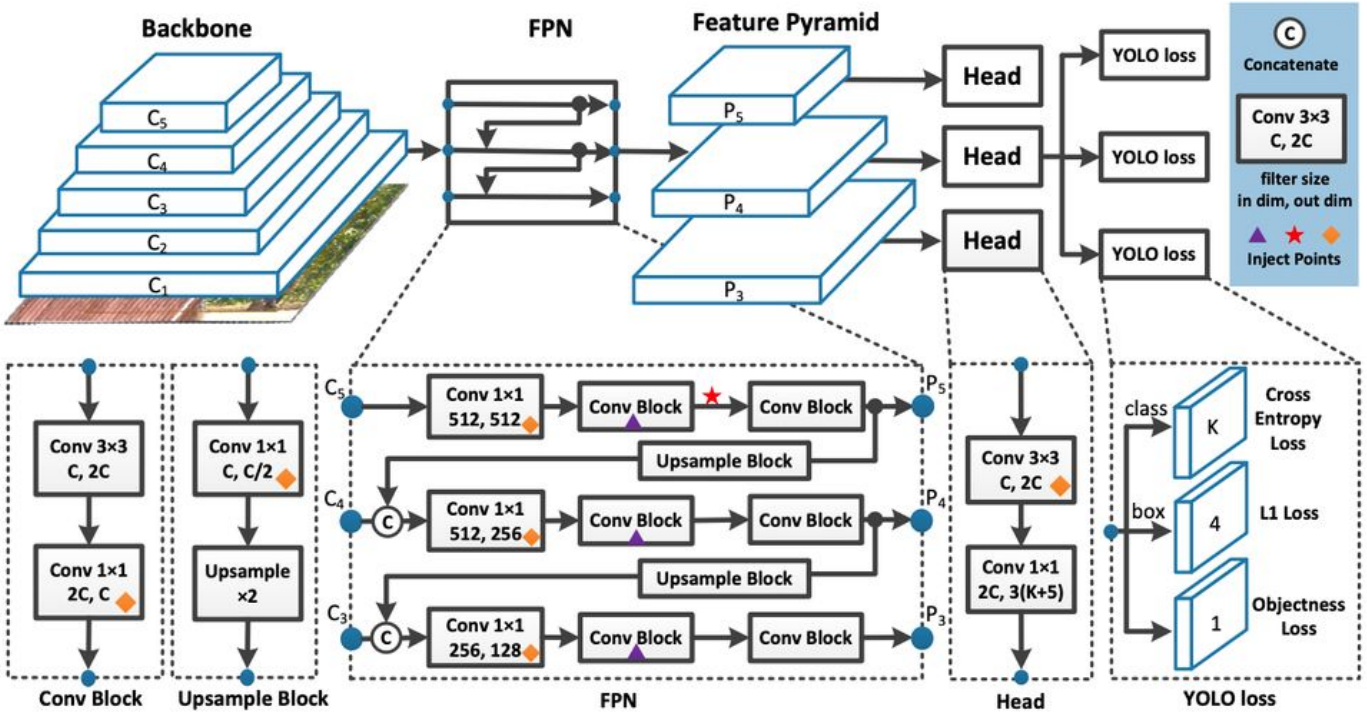
YOLOv1 to YOLOv3 (2016 - 2018)

v4



YOLOv4 to YOLOv7 (2020–2021)

v7



Object Tracking Technologies

They work on predicting object trajectory

DeepSORT (2017): CNN-Based Appearance Embedding

ByteTrack (2022): Associates all detections (high + low confidence)

BoT-SORT (2023): Integrates motion and re-ID optimization

Object detector (e.g., YOLO) → feature extraction → data association → trajectory update

Object Tracking Technologies

They work on predicting object trajectory

FairMOT: Joint Detection and Tracking

CenterTrack: Regression-Based Tracking

TransTrack: Transformer-Based Multi-Object Tracking

Jobs

Assign ID, Track Object, Re Assign ID after Disppear

YOLOv8 to YOLOv12 (2020–2025)

Area Attention Mechanism **Block A**

Residual Efficient Layer Aggregation Network **(RELAN)**

Anchor-Free Detection Head

Decoupled Head **Separate Localization and Classification**

Programmable Gradient Information (PGI) **Improve Gradient Flow**

Generalized Efficient Layer Aggregation Network (GELAN)

Focus more on Computation Improvements

YOLOv8 to YOLOv12 (2020–2025)

Layer Nomenclature

C2f: a Cross-Stage Partial (CSP) style block with 2 convolutions; faster than earlier bottlenecks.

C3: CSP block with 3 convolutions (previous versions of YOLO used C3).

A2C2f: Attention-enhanced C2f block (area-attention + C2f) → sometimes also called “R-ELAN variant”.

YOLOv8 to YOLOv12 (2020–2025)

Putting It All Together: YOLOv12 Pipeline

Input Image (e.g., 640×640) → pre-processing (normalisation, etc).

Backbone: stacks of conv blocks (C2f/C3) → intermediate features → attention-enhanced blocks (A2C2f, Area Attention) → deeper features with large receptive field.

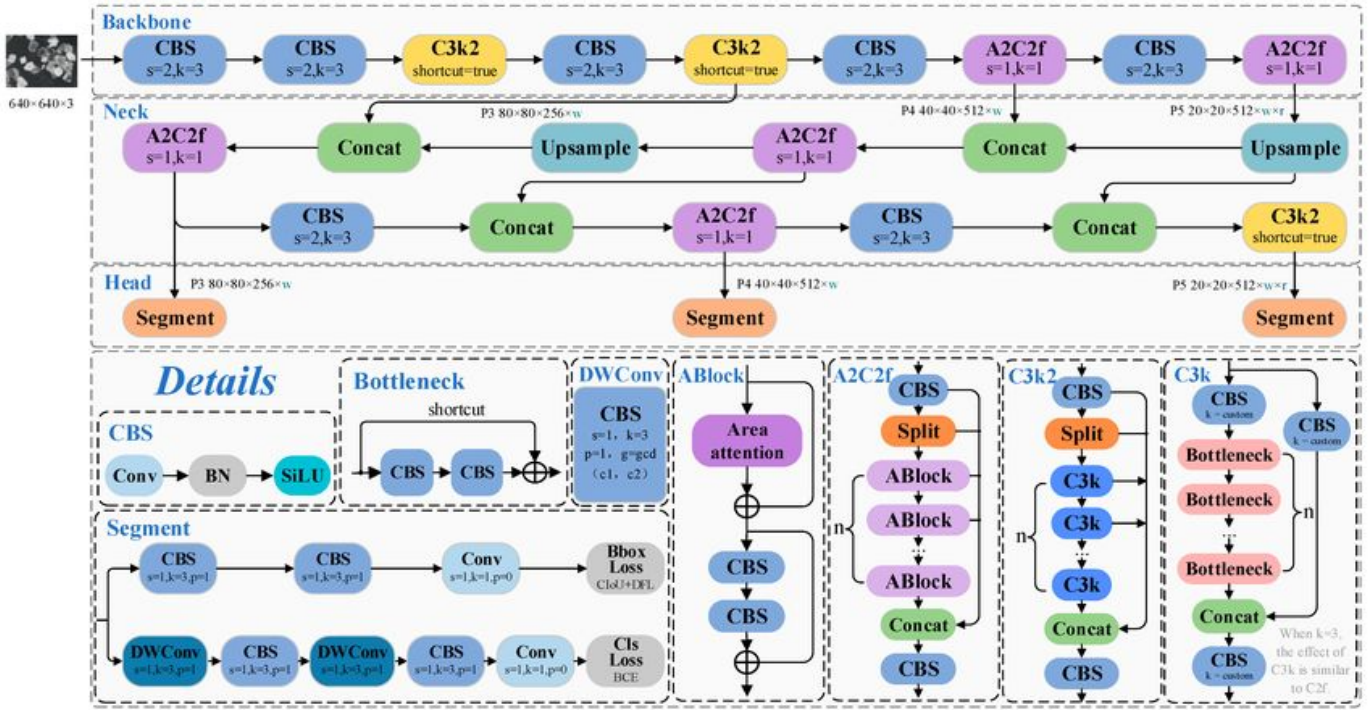
Neck: multi-scale feature maps are fused: upsampling/downsampling, concatenation, blocks like A2C2f used for fusion.

Head: For each scale: prediction of bounding boxes (using regression + classification losses) + possibly segmentation/pose.

Post-processing: Non-Max Suppression (or other methods) to filter, plus multi-task outputs if applicable.

YOLOv8 to YOLOv12 (2020–2025)

v12



Code Walking: YOLO Training

Data Structure

DATA_DIR = '**/content/data**'

IMAGES_DIR = os.path.join(DATA_DIR, '**images**')

LABELS_DIR = os.path.join(DATA_DIR, '**labels**')

Train/Test/Split

SPLIT_RATIO = (0.7, 0.2, 0.1) # train, val, test

Code Walking: YOLO Training

Train/Test/Split

```
# First split: train+val vs test
train_val, test_images = train_test_split( valid_images, test_size=test_ratio,
random_state=42)

# Second split: train vs val
val_size = val_ratio / (train_ratio + val_ratio)
train_images, val_images = train_test_split(  train_val,   test_size=val_size,
    random_state=42)
```

Code Walking: YOLO Training

Create YAML

```
OUTPUT_DIR = os.path.join(DATA_DIR, 'yolo_dataset')
data_yaml = {
    'path': str(Path(OUTPUT_DIR).absolute()),
    'train': 'train/images',    'val': 'val/images',    'test': 'test/images',
    'nc': len(classes),    'names': classes
}
with open(yaml_path, 'w') as f:
    yaml.dump(data_yaml, f, default_flow_style=False)
```

Code Walking: YOLO Training

Download / Load Model

```
from ultralytics import YOLO
model_v8 = YOLO('yolov8n.pt')   # 'n', 's', 'm', 'l', 'x'
# Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt to 'yolov8n.pt
```

Modeling

```
results_v8 = model_v8.train(
data=str(yaml_path),
epochs=EPOCHS,
imgsz=IMG_SIZE,
```

Code Walking: YOLO Training

Modeling

```
        patience=50,      Early Stopping Patience
        save=True,        Saves Model Checkpoints (Logs, Weights)
        project=str(RESULTS_DIR_V8),
        name='candy_detection_v8',  Inside Project this will Appear with Logs, Weights
        exist_ok=True,    Reuse the folder if already Present
        plots=True,       Saves all Line Graphs for training and Validation
        verbose=True      Show Progress in Console
    )
```

Code Walking: YOLO Training

```
└─────candy_detection_v9
      args.yaml
      BoxF1_curve.png
      BoxPR_curve.png
      BoxP_curve.png
      BoxR_curve.png
      confusion_matrix.csv
      confusion_matrix.png
      confusion_matrix_normalized.png
      labels.jpg
      metrics.json
      results.csv
      results.png
      train_batch0.jpg
      train_batch1.jpg
      train_batch2.jpg
      train_batch630.jpg
      train_batch631.jpg
      train_batch632.jpg
      val_batch0_labels.jpg
      val_batch0_pred.jpg
      val_batch1_labels.jpg
      val_batch1_pred.jpg

      └─────weights
            best.onnx
            best.pt
            best.torchscript
            last.pt
```

Code Walking: YOLO Inference CLI

Image

!yolo **predict** model=/content/runs/detect/train/weights/best.pt **source**=/content/image.jpg

Video

!yolo **predict** model=/content/runs/detect/train/weights/best.pt **source**=/content/video.mp4 **imgsz**=640 conf=0.5 **device**=0 **project**=/content/results **name**=video_pred **exist_ok**=True

Code Walking: YOLO + Tracking

Now you can use Save Yolo file for Detection and one of Tracking Algo with it lets SEE

```
from deep_sort_realtime.deepsort_tracker import DeepSort
tracker = DeepSort(
    max_iou_distance=0.7,      Ratio upto Which object is Categorized same
    max_age=30,      Number of Frames upto which Algo look for Same Object  Re-ID
    n_init=3, Number of consecutive detections required before confirming a New Track
    nn_budget=100  Memory size for ReID Network
)
```

Code Walking: YOLO + Tracking

```
model = YOLO('/content/v9/weights/best.pt')

results = model.predict(source=frame, verbose=False)
```

```
for r in results:
    boxes = r.boxes
    for box in boxes:
        x1, y1, x2, y2 = box.xyxy[0].tolist()
        conf = box.conf[0].item()
        cls = int(box.cls[0].item())
        if cls == pig_class_id and conf > 0.75:
            bbox_xywh = [x1, y1, x2 - x1, y2 - y1]
            detections.append((bbox_xywh, conf, cls))
tracked_objects = tracker.update_tracks(detections, frame=frame)
```

Code Walking: YOLO Detection + Segmentation

```
…
model = YOLO("yolo11n-seg.pt")   -seg mean its will give mask as well in Result
…
   results = model(source=..., stream=True)  # generator of Results objects
    for r in results:
        boxes = r.boxes  # Boxes object for bbox outputs
        masks = r.masks  # Masks object for segment masks outputs
        probs = r.probs  # Class probabilities for classification outputs
```

Code Walking: YOLO on Videos

```
results = model.predict(
    source=video_path,
    save=True,
    show=False,
    verbose=False,
    stream=False
)
```

This will give BBx, Mask, and Track Object upto Visibility … Noice

Code Walking: SAM

```
from ultralytics import SAM

…

model = SAM("sam2.1_b.pt")

…

results = model("/content/image.jpg")

…
```

Give BBx, Mask, Unique ID to each Object by Default

Code Walking: SAM Video Tracking

As Initial you have to give BBx for Object/s to SAM to Tack

from ultralytics.models.sam import **SAM2VideoPredictor**

…

results = **yolo_model**(first_frame)

first_frame_boxes = results[0].boxes.xyxy.cpu().numpy()

…

sam_predictor = SAM2VideoPredictor(overrides={"model": "sam2.1_b.pt"})

sam_results = **sam_predictor**(source=video_path, bboxes=[first_frame_boxes])

Code Walking: SAM Video Tracking

sam_results = "frame_id": i, "class_id": cls_ids[j], "confidence": confs[j],

"x1": box[0], "y1": box[1],

"x2": box[2], "y2": box[3]

….

```
for i, res in enumerate(sam_results):
    if hasattr(res, "boxes") and res.boxes is not None:
        boxes = res.boxes.xyxy.cpu().numpy()  # (x1, y1, x2, y2)
        confs = res.boxes.conf.cpu().numpy() if res.boxes.conf is not None else [None]*len(boxes)
        cls_ids = res.boxes.cls.cpu().numpy().astype(int) if res.boxes.cls is not None else [-1]*len(boxes)
```

Thanks

Problems

Short-term Occlusion: Trackers lose the object when it goes behind another object or obstacle.

Long-term Reappearance (Re-ID): Reassigning the correct ID after a long disappearance remains difficult.

Life Span: For How long keep track of Trajectory

Fragmentation: When a single continuous object trajectory splits into multiple shorter tracks.

Drift: Gradual misalignment between predicted track and true position due to motion model inaccuracies.